

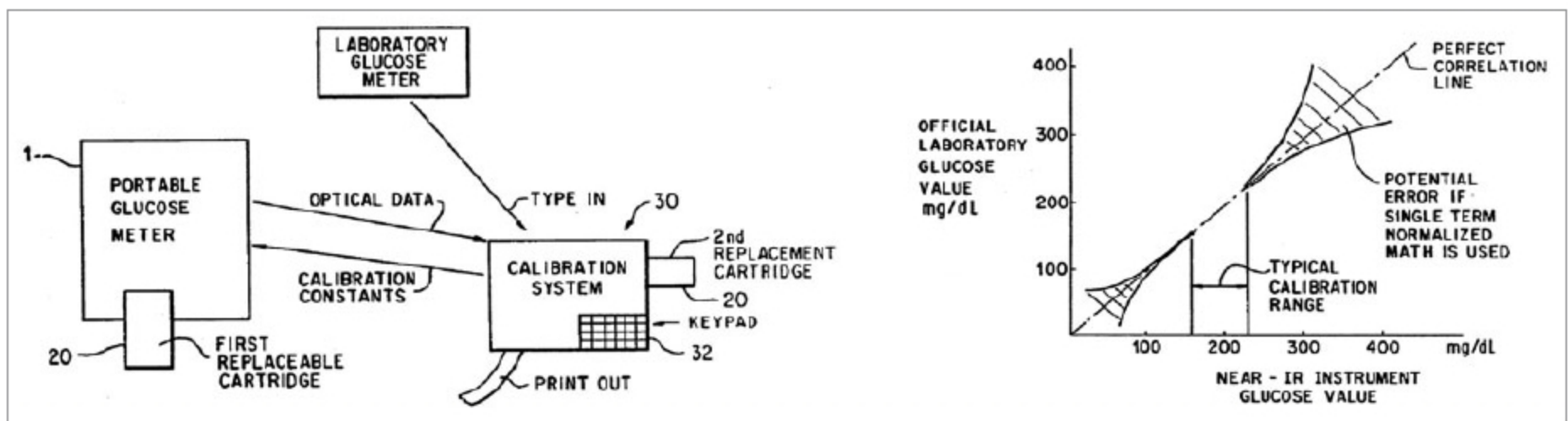


Fig. 4: System calibration and calibration curve of Patent No. 5,077,476

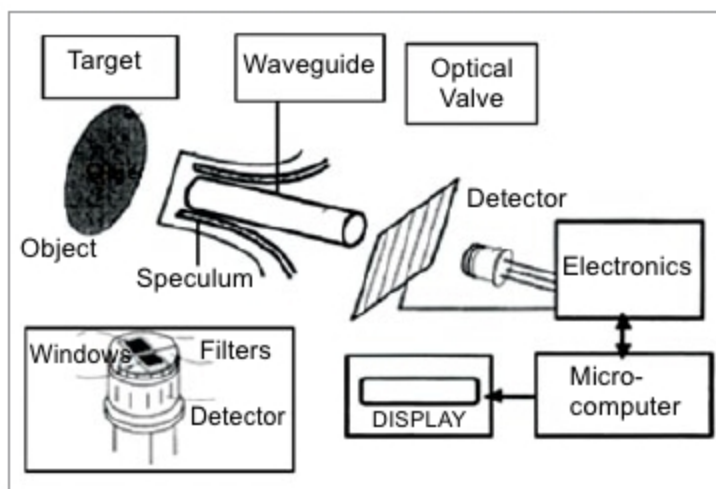


Fig. 5: Block schematic of IR monitoring system

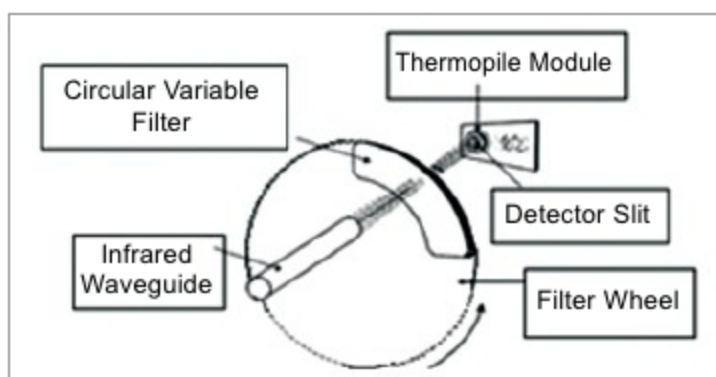


Fig. 6: Block schematic of IR spectro-radiometer instrumentation

Non-invasive measurements and calibration criteria

Calibration of instruments and validation of results. Calibration of instruments and validation of results obtained by non-invasive optical methods under different environmental conditions and used by different patient populations based on different age, sex, size and ethnic origin must be performed. The devices may have to be calibrated for individual users and EGA of data.

Existing glucose measurements. Existing instrumentation requires specificity due to substantial chemical and physical interferences. The devices use multi-variate regression analysis that converts the optical

signal into a glucose concentration (mg/dl) value. Large amounts of data are used to build the glucose model. These must take into consideration concentration range, sampling environment and other factors involved in the analysis.

True glucose value and treatment. The measuring instrument must be designed for accurate glucose concentration (mg/dl). Correlation and clinical interpretation of this value, with respect to true glucose value, is obvious for optimum therapy and diabetes management.

Although non-invasive patented technology has applications for measuring the levels of various substances in the human body, an accurate non-invasive BG measurement technique is the need of time over existing invasive techniques.

Non-invasive measurement technology and instruments. While discussing optical non-invasive BG measurement methods and accuracies, the study renders strategies, non-invasive techniques and their evaluations. Among these methods, some are patented, and some have shown promising accuracies. These are discussed below.

1. In 1991, Robert Rosenthal, Gaithersburg, US, registered Patent No. 5,077,476 for the invention related to non-invasive quantitative measurement of BG instruments and methods. More particularly, this patent provides a handheld BG analysis instrument including a replaceable cartridge con-

taining data concerning a particular patient.

This quantitative analysis instrument measures BG non-invasively by analysing near IR (NIR) energy following interaction with venous or arterial blood, or transmission through blood containing body part. A replaceable cartridge stores data unique to individual patient or user. The instrument is accurate and can be used at home for testing by diabetics.

The analysis instrument comprises a lightweight handheld instrument. This includes a housing unit and a replaceable cartridge, which includes the base unit. Data stored in the replaceable cartridge may include a series of glucose readings and calibration constants, which are custom-calibrated for individual users.

System calibration and results of Patent No. 5,077,476 are shown in Fig. 4.

2. In non-invasive BG monitoring using NIR spectroscopy, In-Ga-As photodiode is used to sense NIR signals, which can also sense other wavelengths of green, red and IR, as their spectral response includes these wavelengths. All variables are amplified, sampled and processed using PSoC5LP. Measurements are communicated through Bluetooth to an Android application.

Light transmitters and receptors use around 1550nm wavelength. This non-invasive NIR spectroscopy BG meter can provide glucose measurements painlessly, without a blood sample, in a few seconds. The device can be easily modified to provide continuous BG and blood oxygen level monitoring of a patient.

Statistical analysis of measure-